

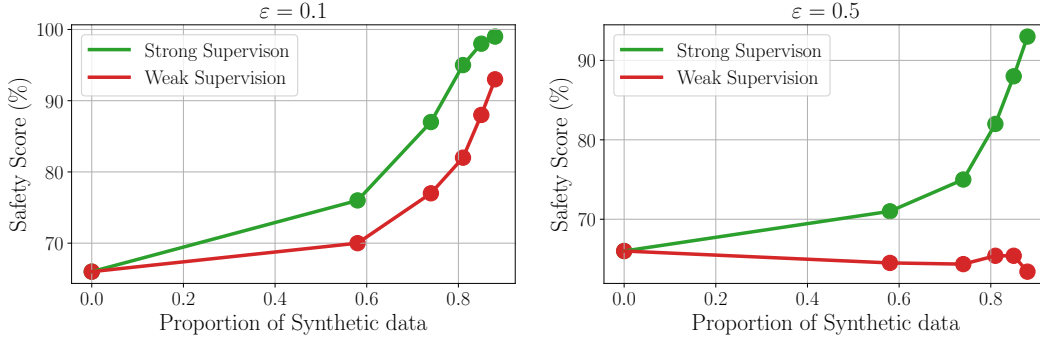


Figure 8: **Results of LLM Safety Alignment:** Strong supervision corresponds to $(\rho, \phi) = (0.2, 0.9)$ and weak supervision to $(\rho, \phi) = (0.5, 0.5)$.

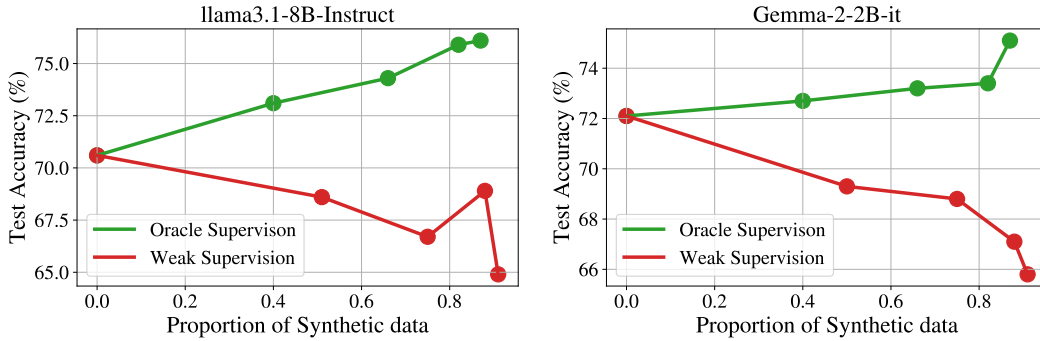


Figure 9: **Results of LLM Q&A Safety Generation:** Evaluation of two LLMs trained as presented in section 5.1 is depicted for both (left) $M = \text{Llama3.1-8B-Instruct}$ and (right) $M = \text{Gemma-2-2B-it}$. The test accuracy is computed over the testing dataset extracted from Ji et al. (2024), with 2.8k Q&A samples. The results shown are the average over 3 runs.

6 CONCLUSION AND LIMITATIONS

In this work, we conducted a comprehensive theoretical and empirical analysis of models trained on a mixture of real and synthetic data with verification. By leveraging random matrix theory, we identified critical factors such as distribution shifts and label noise that significantly impact model performance. Our findings demonstrate that synthetic data can enhance model accuracy under specific conditions, particularly when the generative model is of high quality and the verification process is accurate. Additionally, we extended previous research by showing that performance transitions are smooth rather than sharp when synthetic data is incorporated in high-dimensional settings.

Despite these advancements, our current setting is limited to label verification of synthetic data. Incorporating feature verification represents a promising extension for future research, which could provide further insights into the reliability and effectiveness of synthetic data in model training. Another possible extension of our work is to study distributions beyond the Gaussian model and analyze how higher-order statistics can be incorporated into our current framework.

In conclusion, this work provides a foundational understanding of the conditions under which synthetic data can be beneficial for model training in high-dimensional settings. By integrating both theoretical insights and empirical validations, this study provides new insights into the effective utilization of synthetic data, paving the way for more resilient and performant AI models.

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